International Institute of Information Technology, Bhubaneswar
Department of Computer Science and Engineering

Object Oriented Programming Laboratory

Lab 2 – C++ Programming Assignment (Group B)

Date: 07.08.2026 **Semester:** B.Tech 3rd Semester **Section:** CSE B1

Instructions

- Write all programs in **C++**.
- Use **classes and objects** in every program.
- Implement the required functionality using **member functions**.
- Display the output in a neat and readable format.
- Use appropriate data types and meaningful variable names.

1. **Car Information System**
   Create a class named `Car` to store the details of a car. The class should contain the following data members:

   - Car Number
   - Brand Name
   - Model Year

   Write suitable member functions to:

   1. Accept the car details from the user.
   2. Display the entered details in a formatted manner.

   Hint: *Use separate functions for input and display.*

2. **Square Calculator**
   Create a class named `Square` that stores the side of a square. Write member functions to:

   1. Read the side length.
   2. Calculate the area.
   3. Calculate the perimeter.
   4. Display the results.

   Hint: *Area = Side $\times$ Side, and Perimeter = $4\times$ Side.*

3. **Temperature Converter**
   Create a class named `Temperature`. Write member functions to:

   1. Accept temperature in Celsius.
   2. Convert it into Fahrenheit.
   3. Display both temperatures.

   Hint: *Use the formula:* $F = \frac{9}{5}C + 32$

4. **Hotel Room Booking**
   Design a class named `HotelRoom`. The class should contain:

   - Room Number
   - Guest Name
   - Number of Days Stayed
   - Cost Per Day

   Write member functions to:

   1. Accept booking details.
   2. Calculate the total room rent.
   3. Display the booking details.

   Hint: *Total Rent = Number of Days Stayed* $\times$ *Cost Per Day.*

5. **Mobile Recharge System**
   Create a class named `MobileRecharge`. The class should contain:

   - Mobile Number
   - Customer Name
   - Current Balance

   Write member functions to:

   1. Accept customer details.
   2. Recharge the balance.
   3. Deduct the balance after selecting a recharge plan.
   4. Display the updated balance.

   Hint: *Ensure that the recharge amount does not exceed the available balance.*

6. **Time Addition**
   Create a class named `Time` with two data members: **Hours** and **Minutes**. Write member functions to:

   1. Input two time values.
   2. Add the two time values.
   3. Display the resulting time.

   Hint: *Convert every 60 minutes into 1 hour. Example: 2 hr 45 min + 1 hr 30 min = 4 hr 15 min.*

7. **Movie Ticket Booking**
   Create a class named `MovieTicket`. The class should contain:

   - Movie Name
   - Ticket Price
   - Number of Tickets

   Write member functions to:

   1. Accept booking details.
   2. Calculate the total ticket cost.
   3. Display the booking summary.

   Hint: *Total Cost = Ticket Price $\times$ Number of Tickets.*

8. **Hostel Fee Management**
   Create a class named `HostelFee`. The class should contain:

   - Student Name
   - Hostel ID
   - Monthly Fee
   - Number of Months

   Write member functions to:

   1. Accept student details.
   2. Calculate the total hostel fee.
   3. Apply a late fine of ₹500 if the payment is delayed.
   4. Display the final amount payable.

   Hint: *Use an additional variable to indicate whether the payment is delayed.*

9. **Cricket Score Analyzer**
Create a class named `CricketPlayer`. The class should contain:

- Player Name
- Matches Played
- Total Runs Scored

Write member functions to:

1. Accept player details.
2. Calculate the batting average.
3. Classify the player's performance using the following criteria and display the complete player report:

Batting Average	**Performance**
50 and above	Excellent
35–49.99	Good
20–34.99	Average
Below 20	Poor

Hint: *Batting Average = Total Runs / Matches Played.*

10. **Water Bill Calculator**
Create a class named `WaterBill`. The class should contain:

- Consumer Number
- Consumer Name
- Water Consumption (in litres)

Write member functions to accept the consumer details, calculate the water bill using the following slab rates, and display the complete bill.

Water Consumption	**Rate**
First 500 litres	₹2 per litre
Next 500 litres	₹3 per litre
Above 1000 litres	₹5 per litre

Hint: *Calculate the bill slab-wise instead of multiplying all litres by a single rate. Example for 1200 litres: (500 × ₹2) + (500 × ₹3) + (200 × ₹5) = ₹3500.*